

Use-Case Flow Specification

Use Case: UC-01 — Generate Mock API Endpoints

Primary Actor: API Developer

Supporting Actor: QA Engineer

System Input: OpenAPI 3.0 YAML/JSON specification

Preconditions

- The user has a valid OpenAPI 3.0 YAML or JSON specification available.
- The OpenAPI Mock Server Generator is running and ready to accept the specification.

Postconditions

- Dynamic mock REST endpoints are generated from the specification.
- Generated endpoints use schema-compliant randomized JSON responses.
- Configured response latency is applied to mock responses.

Main Success Scenario

1. API Developer provides an OpenAPI 3.0 YAML/JSON specification to the system.
2. System parses the OpenAPI specification and its schema definitions.
3. System validates that the supplied specification can be processed.
4. System generates dynamic HTTP mock endpoints for the defined API paths and operations.
5. System associates request and response schemas with the generated endpoints.
6. System configures the requested simulated response latency.
7. API Developer or QA Engineer sends a request to a generated mock endpoint.
8. System validates the request against the applicable OpenAPI request schema.
9. System generates a randomized JSON payload that conforms to the defined response schema.
10. System waits for the configured simulated latency and returns the mock response.
11. Use case ends successfully.

Alternate Flow — Invalid OpenAPI Specification

- AF-1. At Step 2 or 3, the system detects that the OpenAPI specification is malformed or cannot be validated.
- AF-2. System rejects the specification and does not generate mock endpoints.
- AF-3. System displays a clear validation error without exposing internal implementation details.
- AF-4. API Developer corrects the specification and submits it again, or cancels the operation.